

Motivations

- Replicate humans' capacity for *intuitive physics*, drawing from neuroscience while omitting explicit priors.
- Accurate physical reasoning in deterministic settings serves as a critical foundation for robust performance in more complex, stochastic environments.
- Our approach is grounded in biological plausibility, extending the predictive coding framework of PredNet.

Loss

$$\mathcal{L}_{total} = \lambda_{loc}\mathcal{L}_{loc} + \lambda_{mask}\mathcal{L}_{mask} + \lambda_{mae}\mathcal{L}_{mae} + \lambda_{vgg}\mathcal{L}_{vgg} + \dots$$

What Pathway

- $\mathcal{L}_{vgg}$: Perceptual loss (VGG16)
- $\mathcal{L}_{mae}$: Weighted MAE loss that amplifies salient object regions ($\gamma=15$)
- $\mathcal{L}_{err}$: Internal error loss

Where Pathway

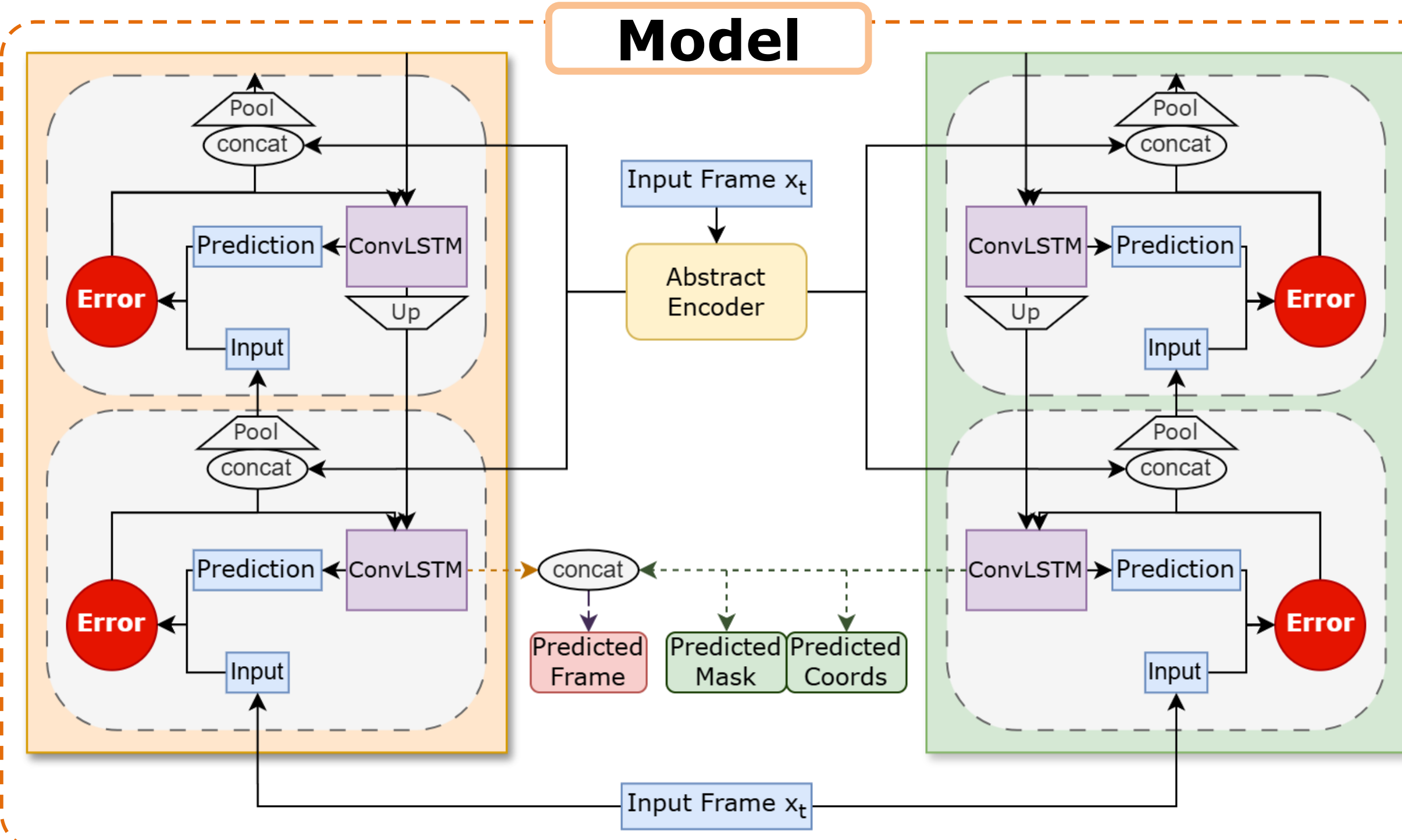
- $\mathcal{L}_{loc}$: Auxiliary location loss (3D coordinates)
- $\mathcal{L}_{mask}$: Dice & Focal loss of segmentation mask
- $\mathcal{L}_{err_loc}$: internal error loss

Findings

- While our model produces high-fidelity images, it exhibits a *reactive one-frame lag* during collisions.
- A special configuration using a heavily-weighted centroid loss ($\mathcal{L}_{centroid}$) anticipated a collision and produced a physically plausible rebound, although at the expensive of fidelity. This indicates competing objectives, which was reinforced by ablation studies.

Methods

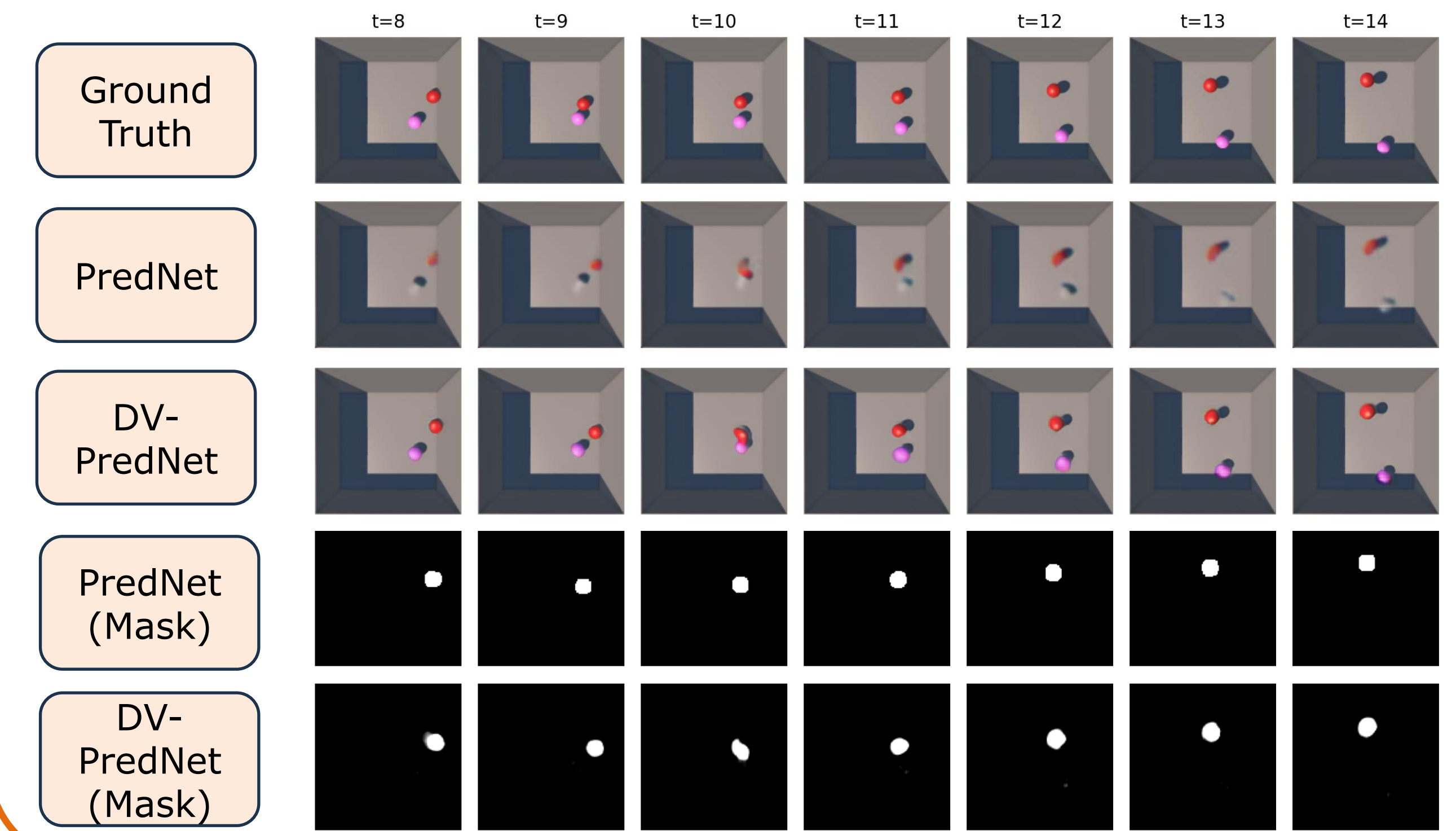
- Inspired by humans' visual processing stream, our model disentangles the learning of object representation (**what stream**) and spatiotemporal dynamics (**where stream**). The **what stream** prioritizes high-fidelity reconstruction of salient objects in the next frame, while the **where stream** predicts the upcoming segmentation mask.



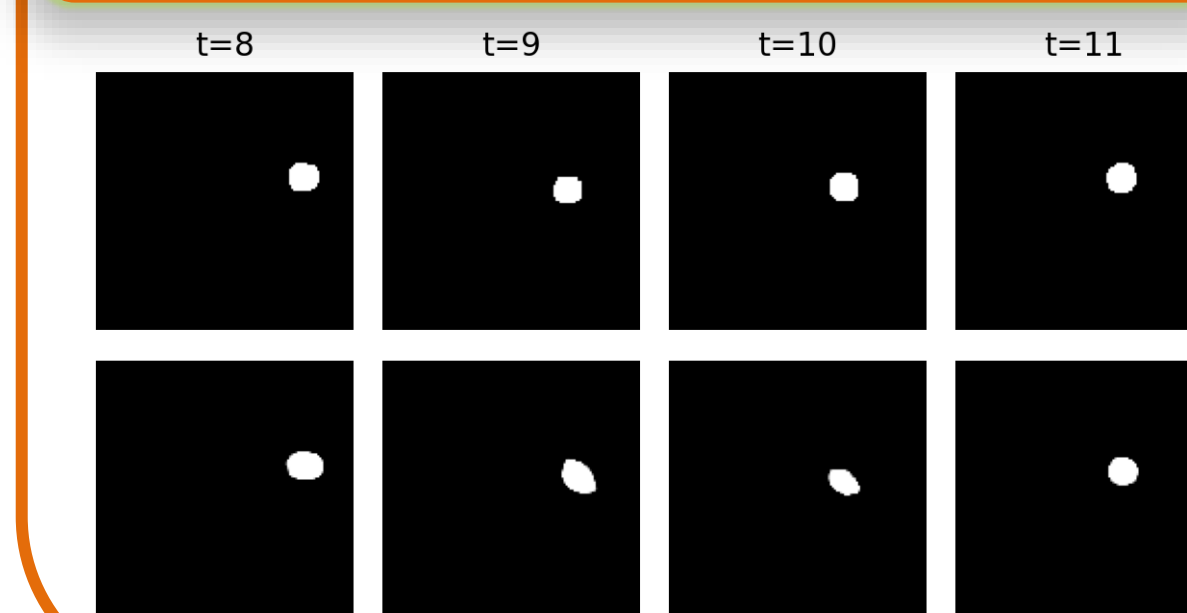
- A *shared encoder* provides both streams with an abstract representation of static scene elements; it is especially important for the **where stream**, which would otherwise lack such information.
- Event-based weighting is used to amplify loss contributions of frames involving collisions ($\gamma=15$).

Results

	Perceptual	SSIM	PSNR
PredNet	0.596	0.979	34.14
DV-PredNet	0.138	0.988	38.59
DV-PredNet ($\mathcal{L}_{centroid}$)	0.159	0.981	35.01



$\mathcal{L}_{centroid}$ Mask Prediction



Conclusions

- While our model improves visual fidelity, we observe competing visual and physical objectives that lead to a reactive, one-frame lag during collisions.

Reference: W. Lotter, G. Kreiman, and D. Cox. Deep predictive coding networks for video prediction and unsupervised learning, 2017. arXiv preprint arXiv:1605.08104.336